



What is Scalability?

This lesson introduces scalability.

We'll cover the following



- What is scalability?
- What is latency?
- Measuring latency
 - Network latency
 - Application latency
- Why is low latency so crucial for online services?

I am pretty sure, being in the software development universe, you've come



contingencies to scale when your app or the platform experiences significant traffic growth?

This chapter is a deep dive into *scalability*. It covers all the frequently asked questions about it, including what does *scalability* mean in the context of web applications and distributed systems?

So, without further ado. Let's get started.



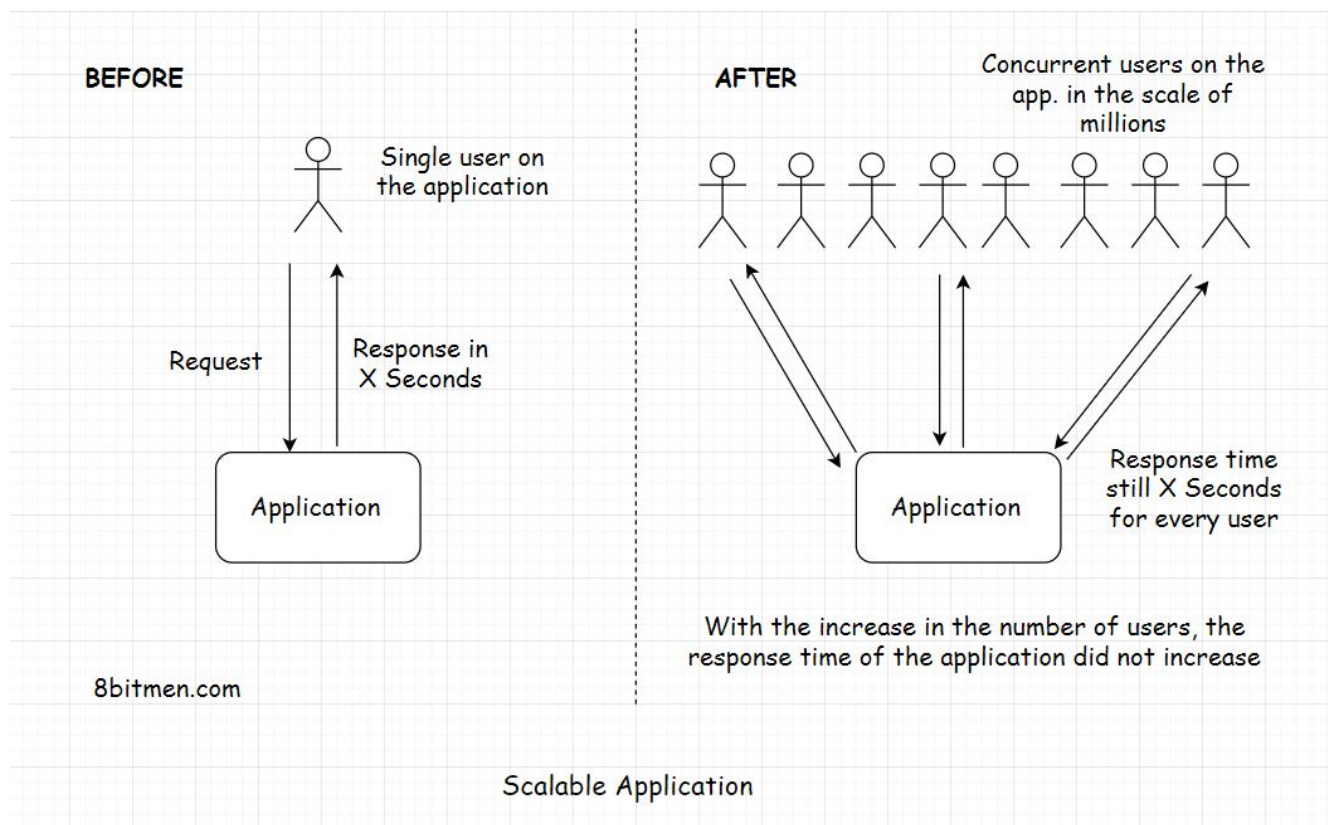
What is scalability?



Scalability means the application's ability to handle and withstand increased workload without sacrificing performance.

For example, if your app takes x seconds to respond to a user request. It should take the same x seconds to respond to each of your app's million concurrent user requests.

The app's back-end infrastructure should not crumble under a load of a million concurrent requests. It should scale well when subjected to a heavy traffic load and maintain the system's latency.



What is latency?

Latency is the time a system takes to respond to a user request. Let's say you send a request to an app to fetch an image and the system takes 2 seconds to respond to your request. The latency of the system is 2 seconds.

Minimum latency is what efficient software systems strive for. No matter how much the traffic load on a system builds up, the latency should not go

up. This is what *scalability* is.

If the latency remains the same, we can say that the application scaled well with the increased load and is highly scalable.

Let's see *scalability* in terms of *Big-O* notation. Ideally, the complexity of a system or an algorithm should be $O(1)$ which is constant time like in a *map* or a *key-value* database.

A program with the complexity of $O(n^2)$ where n is the size of the data set is not scalable. As the size of the data set increases, the system will need more computational power and other resources to process the tasks.

So, how do we measure latency?

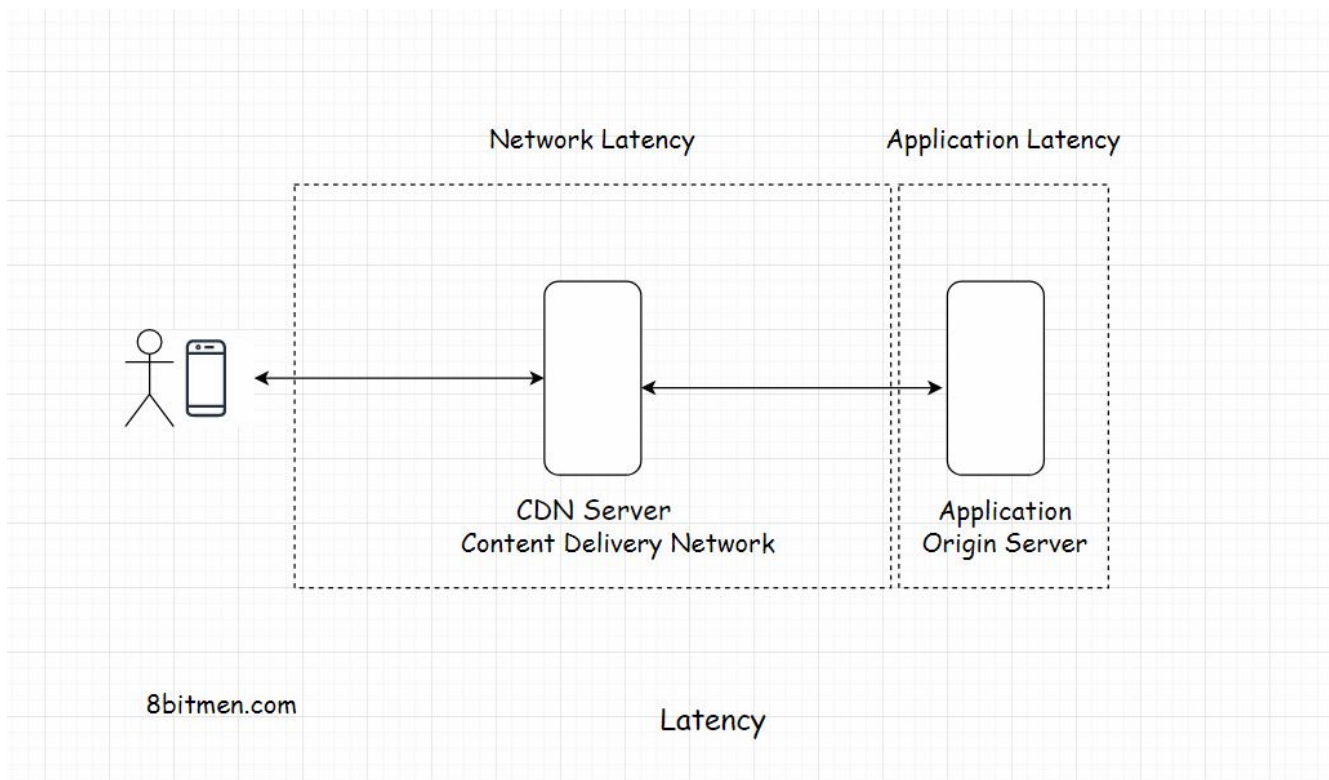
Measuring latency

Latency is measured as the time difference between the action that a user takes on the website and the system's response in reaction to that action. The action can be an event like clicking a button, scrolling down a web page, etc.

This latency is generally divided into two parts:

- Network latency
- Application latency





Network latency

Network latency is the time that the network takes to send a data packet from point *A* to point *B*. The network should be efficient enough to handle the increased traffic load on the website. To cut down the network latency, businesses use a *CDN* (Content Delivery Network) to deploy their servers across the globe as close to the end-user as possible. These close to the user locations are also known as *Edge* locations.

If you wish to understand the *Edge* locations and how apps are deployed in the cloud. Check out my [cloud computing 101 course](#) on my platform.

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After having spent a decade in the industry writing code, I firmly believe that every software engineer should have knowledge of cloud

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computing. It's the present and the future of application development and deployment.



Application latency

Application latency is the time the application takes to process a user request. There are more than a few ways to cut down the application latency. The first step is to run *stress* and *load tests* on the application and scan for the *bottlenecks* that slow down the system as a whole. I'll talk more about it in the upcoming lessons.

Why is low latency so crucial for online services?

Latency plays a significant role in determining if an online business wins or loses a customer. Nobody likes to wait for a response on a website. There is a well-known saying, "*If you want to test a person's patience, give them a slow internet connection.*"

If the visitor gets the response within a stipulated time, great otherwise, they'll bounce off to another website. There is ample market research that concludes high latency in applications is a big factor in customers bouncing off a website. If there is money involved, *zero latency* is what businesses want. Only if this was possible.

Think of *massive multiplayer online* (MMO) games. A slight lag in an in-game event ruins the whole experience. A gamer with a high latency internet connection will have a slow response time despite having the best reaction time of all the players in an arena.

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Algorithmic trading services need to process events within milliseconds.

Fintech companies have dedicated networks to run low-latency trading. Th

